

Verifier-Assisted versus LLM-Only Pre-deployment Network-Change Validation: A Preregistered Paired Study

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Abstract—We measured whether integrating a deterministic pre-deployment verifier with a bounded automated repair budget increases the fraction of intent-driven network-change proposals that pass semantic validation compared with accepting a single LLM candidate. In a preregistered paired design (study id `cnsmp2026-llm-verifier-predeployment-001`) we executed 300 paired intents (100 per difficulty stratum: low/medium/high) using `gpt-5-mini` and `deterministic_netops_validator_v5`. Each paired intent produced one shared_initial generation that was evaluated under two conditions: baseline (accept single shot) and guarded (deterministic validation and ≤ 1 automated repair). Outcomes: baseline 299/300 unflagged passes (0.9966667), guarded 300/300 (1.0); contingency $n_{11}=299, n_{10}=1, n_{01}=0, n_{00}=0$; absolute paired difference = 0.0033333; exact McNemar two-sided $p = 1.0$; 95% paired bootstrap percentile CI = [0.00,0.01] (bootstrap_seed = 1729, B = 10000). Under the supplied synthetic suite and repair budget the single-shot generator was near ceiling, limiting observable deterministic rescue. We archive preregistration, execution, and analysis artifacts and interpret results with respect to estimand conditioning, validator coverage, repair budget, and operational deployment policies.

I. INTRODUCTION

Generative large language models (LLMs) are being applied to network operations (NetOps) tasks such as intent→configuration translation, configuration synthesis, and automated repair. For pre-deployment network changes, semantic correctness properties—sequencing constraints, reachability, and policy preservation—are operationally critical and cannot be assured by superficial syntactic checks alone. A common engineering pattern is to pair generative models with deterministic validators and bounded automated repair (generate→validate→repair) to reduce operational risk. However, the measurable benefit of such guarded pipelines depends on the experimental estimand (what is held fixed), the generator single-shot quality, the validator’s expressiveness, and the permitted repair budget.

We preregistered a paired experiment (study id `cnsmp2026-llm-verifier-predeployment-001`) to quantify the deterministic rescue achievable when the same single LLM candidate is evaluated under two treatments. For each intent the pipeline produced exactly one shared_initial candidate using the declared generator (`gpt-5-mini`) and compared: (a) baseline — accept the single candidate without repair, and (b) guarded — run a deterministic validator (`deterministic_netops_validator_v5`) and allow at most one automated repair which itself is re-validated. The primary research

question: does adding a deterministic verifier plus at most one automated repair increase the probability that a single LLM candidate passes semantic pre-deployment validation compared with accepting that same candidate without repair?

Contributions of this artifact-grounded study: (1) a preregistered paired evaluation isolating deterministic rescue under a shared_initial estimand that conditions on a single LLM draw per intent; (2) a complete execution and analysis bundle including validator traces and the single automated repair transcript archived with the run; and (3) an evidence-grounded interpretation of estimator conditioning, ceiling effects, and operational implications for NetOps pipelines.

II. RELATED WORK

Prior work on LLMs for network management has pursued two complementary directions that frame our study. First, multiple recent surveys and systems reports document active efforts to apply LLMs to NetOps tasks including intent translation, template synthesis, protocol test generation, and zero-touch workflows [<https://openalex.org/W4402742293>; <https://openalex.org/W4411015066>; <https://openalex.org/W4415275309>]. These surveys synthesize a range of prototypes and experimental systems showing that LLMs can produce syntactically plausible device configurations and candidate workflows, but that practical deployment depends critically on downstream validation and environment modeling. For example, PreConfig (a pretrained model for automating configurations) and other synthesis systems demonstrate automation potential while also highlighting dataset and device-scope differences that limit straightforward generalization [<https://openalex.org/W4392875514>; <https://openalex.org/W4411015066>].

Second, a line of systems research specifically proposes and evaluates tool-augmented LLM architectures that interpose external checkers, planners, or verifiers. Planner-centric and ReAct-style frameworks explicitly enable LLMs to call validators or external tools to check semantics and coordinate multi-step changes; these architectures aim to reduce hallucination and logical errors by shifting some checks outside the model [doi:10.1609/aaai.v40i40.40676; <https://openalex.org/W4389518608>]. Empirical system reports in the NetOps space have implemented generate→validate→repair patterns and reported

task-dependent improvements in correctness when validators capture operational invariants: MeshAgent and related systems show that tool augmentation can improve reliability for certain configuration tasks, while program-repair-oriented work demonstrates that combining symbolic fault localization with generative repair has practical value on constrained benchmarks [<https://openalex.org/W4416927413>; <https://openalex.org/W4404240614>].

Work on error detection and mitigation complements these architecture proposals. Detection methods grounded in model uncertainty or semantic measures (for example, entropy-based detectors) and deterministic safety-net designs have been proposed to catch hallucination or logical inconsistencies before deployment; empirical evaluations indicate these detectors can reduce false assurances but are sensitive to detector design and the class of errors targeted [<https://openalex.org/W4399803256>; doi:10.2139/ssrn.6538460]. Similarly, domain-targeted testing work has shown that LLM agents can generate protocol test cases or device-specific actions, but that correctness requires encoding topological and sequencing invariants into validators or test harnesses to catch semantic failures [<https://openalex.org/W4415275309>].

Two methodological gaps in the literature motivate our paired shared_initial design. Many prior evaluations conflate gains from resampling (drawing multiple independent LLM candidates) with gains that arise solely from deterministic validation and repair; permissive generation policies permit the former and thus measure a different operational tradeoff than a single-call budget. Second, while some works evaluate generate→validate→repair pipelines on task-specific suites, fewer public benchmarks and systematic comparisons isolate semantic correctness (reachability, sequencing, forbidden-template preservation) as a first-class evaluation target across difficulty strata [<https://openalex.org/W4411015066>; <https://openalex.org/W4392875514>]. Our study explicitly conditions on a single shared_initial candidate per intent and uses a deterministic validator with an archived, bounded repair operator to provide an operationally meaningful bound on deterministic rescue that complements existing multi-call or planner-guided evaluations.

III. METHODOLOGY

Preregistration and estimand. The study preregistration (cnsmp2026-llm-verifier-predeployment-001; preregistration SHA-256 = 425cb1652354120cf48f686b964723d64c76d0a8abe53cd5b5ad0ea1af71fc7) fixed the primary estimand as the absolute paired difference in the unflagged verifier-pass proportion (guarded minus baseline) computed on 300 paired cases after applying inclusion/exclusion rules. The confirmatory analysis executor paired_binary_analysis_v1 and all inclusion/exclusion rules were frozen prior to the main run. An optional pilot (n = 60) was described in the preregistration for discordance estimation; that pilot was not executed and no post-preregistration power recalculation was performed

(this deviation is recorded in the archived preregistration and Disclosure Statement).

Task generation and synthetic suite. Tasks were produced deterministically by the registered deterministic_netops_task_generator_v5 into a synthetic_topology_suite stratified into low/medium/high difficulty bins by generator metadata (changed_interface_count, dependency_rule_count, required_command_count). Each task manifest entry encodes the full intent, initial and expected final states, required_sequence constraints, preserve_unspecified_state invariants, and protected_interfaces. Representative difficulty patterns include 'shutdown_configure_restore' (medium) and 'paired_link_atomic_mtu_change' (hard). The synthetic suite enforces validator-checkable invariants used for scoring.

Shared_initial protocol and generator. To implement the shared_initial estimand the pipeline produced exactly one shared_initial generation per pair and reused that identical candidate for both baseline and guarded conditions. The declared generator for shared_initial candidate production was gpt-5-mini (execution/model_configuration.json recorded requested_model = "gpt-5-mini", declared_model_version = "gpt-5-mini"). This isolates deterministic validator/repair rescue from gains achievable by independent resampling, verifier-guided generation, or iterative generation.

Validator, scoring, and repair operator. deterministic_netops_validator_v5 executed the preregistered full_intent_constraint_validation rule: deterministic checks for sequencing invariants (must_be_down_for_fields), required_command_count, preservation of unspecified state, topology reachability invariants encoded in the synthetic suite, and forbidden-template protection for protected_interfaces. Scoring was binary: unflagged == pass. The guarded pipeline permitted at most one automated repair attempt (repair budget <=1). The repair routine was a constrained edit operator limited to localized sequencing or insertion edits; it logged a repair provider trace and was applied only when the validator flagged the shared_initial candidate. Candidates remaining flagged after the permitted repair counted as failures for the primary estimand.

Contamination controls, negative controls, and stopping rules. The preregistration included one intentionally broken negative control per complexity stratum and deterministic invariant checks (pre/post reachability matrices). Per preregistration, pairs with both calls failing due to provider/infrastructure outages would be excluded (none occurred). The preregistered stopping rule would halt if >20% dual-call exclusions occurred; this did not trigger.

Execution instrumentation and audit. The confirmatory run used adapter hosted_netops_gvr_v1 under execution_mode = scientific_confirmatory. Execution manifest values (archived): planned_episode_count = 600, completed_episode_count = 600, complete_pair_count = 300, model_calls_used = 301, maximum_model_calls = 600. Provider call auditing recorded hosted_model_call_event_count = 301, cache_miss_count = 301, stage_counts: shared_initial_generation = 300 and repair = 1. Deterministic reconciliation confirms

episode and provider-call accounting and reports `all_deterministic_consistency_checks_passed = true`. These audit values are archived for third-party verification.

Analysis executor and inferential settings. The preregistered analysis executor `paired_binary_analysis_v1` computed paired contingency counts ($n_{11}, n_{10}, n_{01}, n_{00}$), the absolute paired difference (guarded - baseline), an exact two-sided McNemar test, and a paired bootstrap percentile 95% CI. The bootstrap parameters archived with the analysis are `seed = 1729` and `resamples B = 10000`. Sensitivity analyses specified in the preregistration (failure-as-zero and stratum-wise tests with Benjamini-Hochberg FDR control) were run as exploratory artifacts; the confirmatory results reported here follow the primary analysis plan.

IV. RESULTS

Execution integrity and accounting. The preregistered adapter completed the confirmatory run with `completed_episode_count = 600`, yielding `complete_pair_count = 300` paired observations under the `shared_initial` protocol. Provider/audit records show `hosted_model_call_event_count = 301` (300 `shared_initial` generations plus one repair invocation), `cache_miss_count = 301`, and `stage_counts` indicating `shared_initial_generation = 300` and `repair = 1`. Deterministic reconciliation confirmed consistent episode accounting (`all_deterministic_consistency_checks_passed = true`) and the absence of provider/infrastructure failures or excluded pairs.

Primary confirmatory outcomes. The binary scoring produced `baseline_success_count = 299` and `guarded_success_count = 300` (`analysis/condition_summary.csv`), i.e., `baseline = 299/300` (0.9966666667) and `guarded = 300/300` (1.0). The paired contingency table (`analysis/paired_contingency_table.csv`) is $n_{11} = 299$, $n_{10} = 1$, $n_{01} = 0$, $n_{00} = 0$. The preregistered exact two-sided McNemar test (operating on the discordant counts n_{10} and n_{01}) returned $p = 1.0$. The absolute paired improvement (guarded - baseline) equals 0.0033333333333332993. The paired bootstrap percentile 95% confidence interval computed with `seed = 1729` and `B = 10000` is [0.00, 0.01]. Because discordance is minimal ($m = n_{10} + n_{01} = 1$), the exact test has limited sensitivity and the bootstrap CI bounds the plausible absolute improvement to at most one percentage point in this experimental context.

Per-stratum diagnostics. Per the task manifest stratification, archived per-stratum tallies were: low difficulty `baseline = 100/100` and `guarded = 100/100`; medium `baseline = 100/100` and `guarded = 100/100`; high `baseline = 99/100` and `guarded = 100/100`. Thus all discordant evidence is concentrated in the high stratum (1 rescues), while low and medium strata were at ceiling under both conditions. These counts are reported in the archived `condition_summary.csv` and corroborate that the observed discordance derives from a single hard example under the synthetic generator’s difficulty taxonomy.

Repair diagnostic and episode-level trace. The solitary discordant pair invoked the permitted automated repair. Validator

traces (`deterministic_netops_validator_v5`, `validator_version = 5.0`) attribute the initial failure to a sequencing invariant violation (the preregistered `must_be_down_for_fields` rule): the `shared_initial` candidate omitted an explicit ‘admin down’ step prior to a VLAN/MTU change required by the intent. The constrained repair operator performed a single localized insertion of the missing sequencing command, after which the deterministic validator reported `valid == true` for the corrected candidate. The repair therefore produced a deterministic rescue with one additional model/repair stage call. The repair provider trace and the validator’s before/after traces for that episode are archived; the manuscript conveys the semantic summary rather than reproducing verbatim provider logs in the body.

Contamination, missingness, and negative controls. The preregistered contamination checks produced an empty `contamination_summary.csv` (no contamination flags). The missingness summary reports zero failed or unscorable episodes and zero excluded pairs under preregistered rules (`analysis/missingness_summary.csv`). Negative controls included by the preregistration did not cause unexpected pass behavior in the main run; any negative-control anomalies would have been treated as contamination per the preregistration and are not present in the archived run.

Cost and operational footprint. The guarded pipeline’s deterministic rescue required a single additional repair stage call across 300 pairs (`model_calls_used = 301`), indicating that bounded automated repair under the chosen edit operator imposes minimal extra model-call cost in this run. The empirical `repair_stage_count = 1` (analysis artifact) quantifies the low marginal runtime/model cost for deterministic rescue under the chosen repair budget and edit operator design.

Statistical interpretation and limitations of inference. The confirmatory inferential outcome ($p = 1.0$) primarily reflects the paucity of discordant pairs under the `shared_initial` estimand and chosen synthetic suite. Exact McNemar inference depends only on n_{10} and n_{01} ; with $n_{10} = 1$ and $n_{01} = 0$ the two-sided exact p equals 1.0, indicating no statistically detectable difference under conventional two-sided testing in this sample. The paired bootstrap CI (`seed = 1729`, `B = 10000`) provides a complementary bound for effect size: the absolute paired improvement is plausibly between 0 and 0.01 given observed data and the preregistered estimand. These inferential statements apply strictly to the `shared_initial` conditioning (one LLM draw reused across conditions), the limited repair budget (≤ 1 constrained edit), the synthetic task distribution, and the declared generator (`gpt-5-mini`) and validator (`deterministic_netops_validator_v5`).

Practical diagnostic takeaway. Under operational policies that restrict model calls per change request, the guarded `generate→validate→repair` pattern produced a deterministic rescue in one hard example with negligible additional model-call expenditure in this synthetic suite. However, because baseline single-shot generation was near ceiling (299/300), the marginal observable benefit of bounded deterministic repair was correspondingly small in absolute terms. The archived per-episode validator traces and repair transcript

enable forensic inspection to inform whether richer repair operators, iterative repair budgets, or alternative estimand conditioning (e.g., allowing resampling) would be expected to materially increase rescue rates in different deployment regimes.

V. DISCUSSION

We expand the discussion with technical diagnostics, artifact-grounded interpretation, and operational guidance that follow directly from the archived execution and analysis artifacts.

1) Concrete diagnostics from archived artifacts. Audit fields in the execution manifest quantify cost and intervention footprint: `model_calls_used` = 301 (300 `shared_initial` + 1 repair), `completed_episode_count` = 600, and `repair_stage_count` = 1. The `paired_contingency_table.csv` records $n_{11} = 299$, $n_{10} = 1$, $n_{01} = 0$, $n_{00} = 0$ and the per-stratum summary shows baseline/guarded counts of (100/100, 100/100, 99/100 vs. 100/100 across low/medium/high). The deterministic validator (`deterministic_netops_validator_v5`) applied the preregistered `full_intent_constraint_validation` checks (sequencing invariants, `required_command_count`, `preserve_unspecified_state`, and topology reachability invariants encoded in the synthetic suite). The single rescued failure was a sequencing omission (`must_be_down_for_fields`) corrected by the constrained repair operator; the repair consumed the single extra model call recorded. These concrete, archived diagnostics show exactly how often the bounded repair operator was invoked and what it corrected in the run.

2) Estimand conditioning and what the results mean operationally. The `shared_initial` estimand conditions on a single LLM draw per intent and therefore measures only deterministic verifier/repair rescue applied to that same candidate. Practically, this corresponds to operational policies that limit model calls for cost, latency, or governance reasons. Under such policies the guarded pipeline can only correct defects that are detectable and fixable by a bounded, localized edit applied to the same candidate. In our run, that policy permitted one deterministic rescue (sequencing insertion) at very low extra model cost (one additional call). By contrast, operational pipelines that allow resampling, verifier-guided regeneration, or interactive steering change the estimand: they can exploit generator variance and typically achieve higher pass rates, but at the cost of additional model calls and different governance tradeoffs. Evaluations must therefore align the estimand with the intended deployment policy; the archived artifacts provide a reproducible bound for the single-call policy case.

3) Ceiling effects, pilot utility, and experiment design lessons. The preregistration targeted a detectable paired difference of 10 percentage points with power ≈ 0.8 conditioned on pilot discordance estimates. The optional pilot ($n = 60$) described in the preregistration was not executed; as archived, the main run encountered a near-ceiling baseline (299/300) producing only one discordant observation and correspondingly low sensitivity to small absolute differences. This concrete outcome underscores a methodological lesson:

when generator single-shot performance is uncertain, a small calibration pilot is efficient and often necessary to estimate discordance rates and select an appropriate confirmatory N. The archived design and the omitted pilot are documented in preregistration artifacts and should guide follow-ups: run a small pilot to estimate discordance, then freeze confirmatory N accordingly to avoid underpowered confirmatory runs in ceiling conditions.

4) Validator expressiveness and failure-mode dependence. `Deterministic_netops_validator_v5` enacted a specific set of checks; on our synthetic suite the dominant observed failure class was sequencing omissions. Other potential failure classes—topology reachability violations, ACL misconfigurations, or forbidden-template edits—were not observed at scale here. This demonstrates that a guarded pipeline’s marginal utility depends directly on the joint distribution of generator error types and validator coverage: if failure mass concentrates in classes the validator detects and the repair operator can fix, bounded deterministic rescue is effective; if failures are in classes the validator cannot detect (or repair operator cannot safely edit), the guarded pipeline provides mostly detection rather than automated rescue. The archived `contamination_summary.csv` (empty) and the negative controls embedded in the task manifests provide concrete, testable evidence that the synthetic suite exercised a limited set of failure modes for this run.

5) Repair budget tradeoffs and operational cost accounting. The run’s provider audit shows that each automated repair attempt corresponds to an extra hosted model call (`repair_stage_count` = 1 consumed one call). Therefore, scaling the repair budget or permitting iterative repair increases model calls (and thus cost/latency) approximately linearly per allowed repair round under the same repair implementation. Organizations must trade off budgeted model calls per change against expected rescue yield conditioned on anticipated failure modes. The archived provider auditing fields (`hosted_model_call_event_count`, `cache_miss_count`) make this tradeoff explicit and reproducible for cost modeling.

6) Reproducibility, artifact utility, and forensic analysis. The run archives the full task manifests, `shared_initial` responses, validator traces, and the single repair provider trace for the rescued episode. These artifacts enable forensic inspection of exact edits, the validator state transitions (`validation_before/validation_after`), and the deterministic reconciliation records (`all_deterministic_consistency_checks_passed` = true). For practitioners, the availability of these low-level artifacts supports independent re-scoring, alternative validator checks, and ablation analyses (e.g., applying a different repair operator or richer validator to the same `shared_initial` candidates).

7) Threats to validity and how they map to artifact evidence. Internal validity: provider nondeterminism was low (`hosted_model_call_event_count` = 301, failed calls = 0); execution accounting artifacts show no technical exclusions. Construct validity: our binary unflagged/pass scoring depends on validator coverage as instantiated by determinis-

tic_netops_validator_v5; the synthetic task generator encoded domain invariants used by the validator, which tightens construct alignment but limits generality to other validator definitions. External validity: results are limited to gpt-5-mini and the supplied synthetic_topology_suite; generalizing to different LLMs, richer validators, or real operational networks requires follow-up runs using the same preregistration discipline. Conclusion validity: the near-ceiling baseline produced sparse discordance, constraining inferential power for small effects; the paired bootstrap CI and exact McNemar p documented in analysis artifacts quantify that limitation.

8) Practical recommendations grounded in archived evidence. (a) Align experimental estimands with deployment policies: if budgets constrain calls, evaluate shared_initial rescue; if resampling is allowed, design a different preregistration capturing resampling gains. (b) Run a small pilot to estimate discordance and calibrate confirmatory N; the preregistered but unexecuted pilot in our artifacts illustrates the cost of skipping calibration. (c) Make validator coverage explicit and test it with negative controls targeting diverse failure classes; archived negative controls and contamination artifacts support this practice. (d) Treat repair budget as a tunable parameter with explicit cost accounting: use provider audit fields to model marginal cost per repair call.

Taken together, these artifact-grounded diagnostics and recommendations show how a reproducible pairing of generator, verifier, and constrained repair can be quantitatively audited and how evaluators should select estimands, pilots, and validator tests to obtain operationally meaningful conclusions.

VI. LIMITATIONS

- Synthetic benchmark: tasks and topologies were generated by the preregistered deterministic task generator; real-world heterogeneity may expose different failure modes and increase verifier marginal utility.
- Single model and validator: only gpt-5-mini and deterministic_netops_validator_v5 were evaluated; results may not generalize to other LLMs or richer/diverse validators.
- Ceiling effect: baseline single-shot performance was near 100% on this suite (299/300), producing limited discordance and reduced power to detect small absolute improvements relative to larger pre-specified targets.
- Pilot not executed: the preregistered pilot intended for discordance estimation and power calibration was not run; no post-preregistration power recalculation was performed.
- Estimand conditioning: the shared_initial protocol conditions the estimand on a single LLM candidate and therefore does not measure verifier benefit that would arise from allowing independent alternative LLM candidates per condition (resampling or iterative generation).
- Repair budget: the guarded condition permitted at most one automated repair per pair; larger or iterative repair strategies may yield different outcomes.

VII. CONCLUSION

We executed a preregistered, artifact-archived paired comparison between a verifier-assisted pipeline (deterministic validator + ≤ 1 automated repair) and accepting a single-shot LLM candidate under the shared_initial estimand on 300 synthetic NetOps intents using gpt-5-mini and deterministic_netops_validator_v5. Baseline single-shot generation produced 299/300 unflagged verifier passes and the guarded pipeline 300/300, yielding an absolute paired improvement of 0.00333 (exact McNemar two-sided $p = 1.0$; 95% bootstrap CI [0.00,0.01]). Deterministic validator+repair rescued a single sequencing omission under the bounded repair budget. The near-ceiling baseline limits inferential sensitivity to small absolute improvements under the shared_initial estimand. We release full preregistration, execution, and analysis artifacts to support reproducibility and targeted follow-up studies that vary estimand conditioning, model strength, task distribution, or repair budget.

DISCLOSURE STATEMENT

This experiment and manuscript were produced by an autonomous CNSM Agentic-AI research run executed under the archived initial master prompt (provenance/master_prompt.txt, SHA-256 = 1872df1e1805d2d96940456ca016bd665d1d5196add77f5acdf1582b39b15ba). Human role: the human Research Director selected the topic family (Generative AI and LLMs for NetOps) and approved pre-lock infrastructure; all literature search after the autonomy lock, autonomous study selection, preregistration (study id cnsmp2026-llm-verifier-predeployment-001), execution, analysis, manuscript drafting, AI peer review, and final compilation were performed autonomously under the locked master prompt. Models and orchestration: the declared generation model used was gpt-5-mini (execution/model_configuration.json declares requested_model = "gpt-5-mini"); adapter/orchestration framework: hosted_netops_gvr_v1. Validator and tooling: deterministic_netops_validator_v5 performed semantic and syntactic validation according to the preregistered full_intent_constraint_validation rule; the guarded condition permitted at most one automated repair call per pair implemented as a constrained edit operator. Preregistration: the confirmatory design, estimand, analysis executor, exclusion/missingness rules, and multiplicity plan were frozen prior to the main run and archived with the study id (preregistration SHA-256 = 425cb1652354120cf48f686b964723d64c76d0a8abe53cd5b5ad0eaa1af71fc7). Execution summary (archived): planned_episode_count = 600, completed_episode_count = 600, complete_pair_count = 300, model_calls_used = 301; provider audit: hosted_model_call_event_count = 301, stage_counts show shared_initial_generation = 300 and repair = 1. Pilot: the preregistration described an optional pilot ($n = 60$) for discordance estimation and power calibration; that pilot was not executed and no post-preregistration recalculation of required sample size was performed. Deviations: no

post-preregistration changes to the scientific design were made; no pairs were excluded. Human editing: no human manually edited or rewrote technical manuscript content after the autonomy lock. Artifact availability: all raw outputs, logs, task manifests, validator traces, repair provider traces, and analysis artifacts referenced in this manuscript are archived in the run bundle associated with the study id. The archived run bundle contains the low-level repair provider trace documenting the single automated repair and subsequent validation pass; this manuscript reports a concise semantic summary of that repair in Results rather than reproducing full verbatim provider transcripts in the body.

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